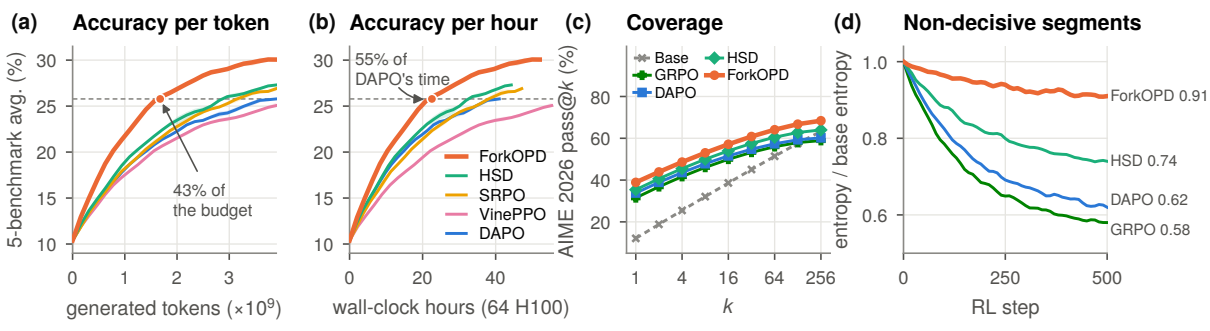
(a)
Accuracy per token
5-benchmark avg. (%)
30
25
20
15
10
0
1
2
3
generated tokens (×10^9)
43% of
the budget
(b)
Accuracy per hour
55% of
DAPO's time
ForkOPD
HSD
SRPO
VinePPO
DAPO
0
20
40
wall-clock hours (64 H100)
(c)
Coverage
AIME 2026 pass@k (%)
Base
HSD
GRPO
ForkOPD
DAPO
80
60
40
20
1
4
16
64
256
k
(d)
Non-decisive segments
entropy / base entropy
1.0
0.8
0.6
ForkOPD 0.91
HSD 0.74
DAPO 0.62
GRPO 0.58
0
250
500
RL step